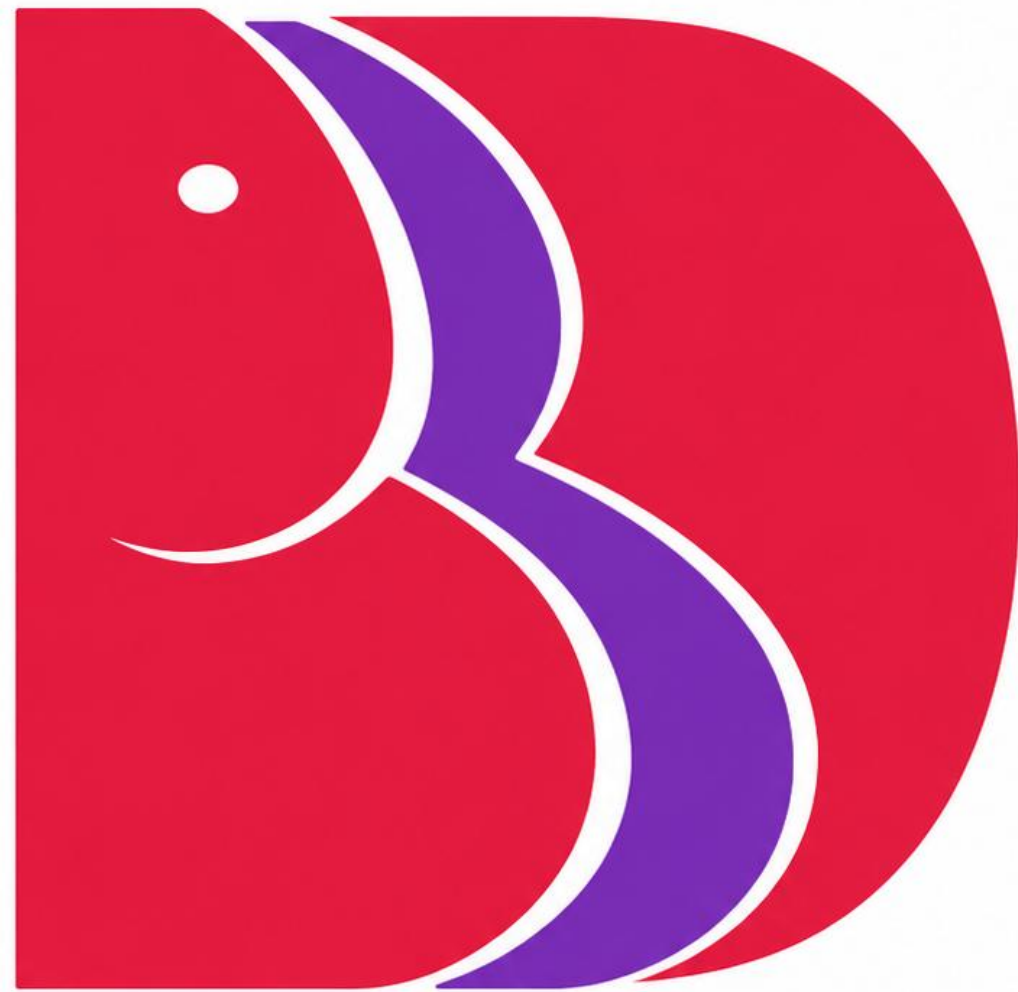


SYLLABUS

Semester 5



**BBD
UNIVERSITY**

Babu Banarasi Das University
B Tech. CSE-AI

ARTIFICIAL NEURAL NETWORKS – NAI4502

Semester 5 Syllabus

MODULE 1

Introduction

Human Brain, Neural Network, Models of a Neuron, Neural Networks viewed as Directed Graphs, Network Architectures, Knowledge Representation, Artificial Intelligence and Neural Networks, Error Correction Learning, Memory Based Learning, Hebbian Learning, Competitive Learning, Boltzmann Learning, Credit Assignment Problem, Memory, Adaption, Statistical Nature of the Learning Process.

MODULE 2

Perceptron

Adaptive Filtering Problem, Unconstrained Organization Techniques, Linear Least Square Filters, Least Mean Square Algorithm, Learning Curves, Learning Rate Annealing Techniques, Perceptron-Convergence Theorem, Relation Between Perceptron and Bayes Classifier for a Gaussian Environment, Multilayer Perceptron, Back Propagation Algorithm, XOR Problem, Heuristics, Output Representation and Decision Rule, Computer Experiment, Feature Detection.

MODULE 3

Back Propagation

Back Propagation and Differentiation, Hessian Matrix, Generalization, Cross Validation, Network Pruning Techniques, Virtues and Limitations of Back Propagation Learning, Accelerated Convergence, Supervised Learning.

CONCEPTS OF DATA SCIENCE WITH PYTHON – NAI4501

Semester 5 Syllabus

MODULE 1

Introduction to Data Science

Case for data science, data science classification, data science algorithms; Data Science Process, prior Knowledge, Data Preparation, Modeling, Application, Knowledge; Data Exploration, Objectives of data Exploration; Datasets, Descriptive Statistics; Data Visualization, Roadmap for data exploration.

MODULE 2

Data Manipulation and Visualization

Introduction, Forms of data graphics, Visualizing categories, Design principles, Visualizing locations, Interaction / animation, visualizing locations, visualizing time, Building and using metrics, visualizing multivariate displays, Perception principles and presenting data graphics, Visualizing multivariate displays.

MODULE 3

Python for Data Science

Introduction to NumPy, Pandas and Matplotlib, How to Import NumPy module, What is data manipulation using Pandas library? Series object in Pandas, DataFrame in Pandas, Loading and handling data with Pandas, Introduction to Matplotlib, Using Matplotlib for plotting graphs and charts like Scatter, Bar, Pie, Line, Histogram and more.

AUTOMATA THEORY AND FORMAL LANGUAGES – NCS4504

Semester 5 Syllabus

MODULE 1

Fundamentals: *Formal Languages, Strings, Alphabets, Languages, Chomsky hierarchy of languages.*

Finite Automata: *Introduction to Finite State Machine, Acceptance of strings and languages, Deterministic Finite Automaton (DFA) and Non-deterministic Finite Automaton (NFA), Equivalence of NFA and DFA – Equivalence of NDFAs with and without ϵ -moves, Minimization of Finite Automata, Equivalence between two DFA's, Finite Automata with output – Moore and Mealy Machines, Conversion of Moore to Mealy and Mealy to Moore.*

MODULE 2

Regular Languages: *Regular expressions, Identity rules, Conversion of a given regular expression into a finite automaton, Conversion of finite automata into a regular expression, Pumping lemma for regular sets, Closure properties of regular sets.*

Context Free Grammars: *Context free grammars and languages, Derivation trees, Leftmost and rightmost derivation of strings and Sentential forms, Ambiguity, Left recursion and left factoring in context free grammars, Minimization of context free grammars, Normal forms for context free grammars, Chomsky normal form, Greibach normal form, Pumping lemma for Context Free Languages, Closure and decision properties of context free languages.*

MODULE 3

Pushdown Automata: *Introduction to Pushdown Automata, Acceptance of context free languages, Acceptance by final state and acceptance by empty state and its equivalence, Equivalence of context free grammars and pushdown automata, Inter-conversion.*

MODULE 4

Turing Machine (TM): *Problems That Computers Cannot Solve, The Turing Machine, Programming Techniques for Turing Machines, Extensions to the Basic Turing Machine, Restricted Turing Machines, Turing Machines and Completeness, Definition of Post's Correspondence Problem, A Language That Is Not Recursively Enumerable, An Undecidable Problem That Is RE, Context Sensitive Languages and Chomsky Hierarchy, Other Undecidable Problems.*

COMPUTER NETWORKS – NCS4503

Semester 5 Syllabus

MODULE 1

Introduction: Network objectives and applications, network structure and architecture, OSI reference model, network services, network standardization, examples of network, TCP/IP model.

Physical Layer: Fundamentals of data communication, transmission media, analog transmission, digital transmission, switching, ISDN, terminal handling, Broadcast channels and medium access, LAN protocols.

MODULE 2

Data Link Layer and Network Layer

Data Link Layer: Design issues, error detection and corrections; elementary data link protocols, sliding window protocols, Examples.

Network Layer: Design issues, routing algorithms, congestion control, internetworking, Examples. CSMA with collision detection, collision free protocols, IEEE standard 802 for LANs, comparison of LANs, Fiber optic network and FDDI.

MODULE 3

Transport, Session and Presentation Layer

Transport Layer: Design Issues, connection management, example of a simple transport protocol.

Session Layer: Design issues, remote procedure call, examples.

Presentation Layer: Design issues, data compression and encryption, network security and privacy.

Application Layer: Design issues, File transfer and file access, electronic mail, virtual terminals, other applications, Case study based on available network software.

ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE – NVC4501

Semester 5 Syllabus

MODULE 1

Introduction to Traditional Knowledge

Introduction to Indian Traditional Knowledge: Understanding the concept and significance of Indian Traditional Knowledge, historical background, and evolution of traditional knowledge in India.

Intellectual Property Rights (IPR): Overview of Intellectual Property Rights and its importance in the context of traditional knowledge. ***Different types of IPRs:*** Copyright, Trademarks, Patents, and Geographical Indications.

Traditional Knowledge and Traditional Cultural Expressions (TCEs): Introduction to Traditional Cultural Expressions and the challenges in their protection. Examination of international frameworks like the WIPO Intergovernmental Committee on Intellectual Property and Genetic Resources, Traditional Knowledge, and Folklore.

Traditional Knowledge and Traditional Ecological Knowledge (TEK): Understanding the relationship between traditional knowledge and traditional ecological knowledge. Analysis of the role of TEK in environmental conservation and sustainable development.

MODULE 2

Traditional Knowledge and IPR Laws in India

Traditional Knowledge and IPR Laws: Study of the legal framework for the protection of traditional knowledge in India. Examination of relevant laws and regulations such as the Traditional Knowledge Digital Library (TKDL), Traditional Knowledge and Patent Law. Understanding the challenges and issues surrounding the patenting of traditional knowledge. Analysis of case studies highlighting the controversies and debates in the field.

Traditional Knowledge and Copyright Law: Exploring the relationship between traditional knowledge and copyright law. Discussion on the issues of cultural appropriation and protection of traditional expressions.

Traditional Knowledge and Geographical Indications (GI): Overview of Geographical Indications and their significance in protecting traditional knowledge. Case studies on the successful registration and protection of traditional products and practices.

Traditional Knowledge, IPR, and the Future: Analysis of current trends and future prospects for the protection and preservation of Indian traditional knowledge. Examination of emerging issues such as digital platforms and traditional knowledge dissemination.

ENGINEERING AND MANAGERIAL ECONOMICS – NHS4501

Semester 5 Syllabus

MODULE 1

Introduction and Basic Concepts

Introduction: Meaning, nature and scope of Economics, Role of Engineering in Economics, Managerial economics and its scope in Engineering perspective.

Demand Analysis, Law of Demand, Determinants of demand, Elasticity of Demand – Price, Income and Cross elasticity, Meaning, Significance and methods of demand forecasting, Law of supply, Factors affecting supply, Elasticity of supply.

MODULE 2

Production Function & Cost Concept

Production Functions, Concept of Iso-quants, Law of Variable Proportion and Law of Return to Scale, Economies & Diseconomies of Scale.

Cost Concepts: An overview of Short run and Long run cost curve – Fixed cost, Variable cost, Average cost, Marginal cost and Opportunity cost.

MODULE 3

Market Structure

Perfect Competition, Imperfect Competition, Monopoly, Monopolistic, Oligopoly, Duopoly, Sorbent features of Price Discrimination.

National Income, Inflation and Business Cycle: Concept of National Income and Measurement, Economic Indicators, Meaning of Inflation, Types, Causes and Prevention methods, Phases of Business Cycle.